In space, surface tension always forces a small blob of liquid to form a sphere. Whenever you see a small solid object that is suspiciously spherical, you can assume it formed in a molten state. If the blob has very high mass, then it could be composed of almost anything and gravity will ensure that it forms a sphere.

Big and massive blobs of gas in the galaxy can coalesce to form near-perfect, gaseous spheres called stars. But if a star finds itself orbiting too close to another object whose gravity is significant, the spherical shape can be distorted as its material gets stripped away. By "too close," I mean too close to the object's Roche lobe—named for the mid-nineteenth-century mathematician Édouard Roche, who made detailed studies of gravity fields in the vicinity of double stars. The Roche lobe is a theoretical, dumbbell-shaped, bulbous, double envelope that surrounds any two objects in mutual orbit. If gaseous material from one object passes out of its own envelope, then the material will fall toward the second object. This occurrence is common among binary stars when one of them swells to become a red giant and overfills its Roche lobe. The red giant distorts into a distinctly non-spherical shape that resembles an elongated Hershey's kiss. Moreover, every now and then, one of the two stars is a black hole, whose location is rendered visible by the flaying of its binary companion. The spiraling gas, after having passed from the giant across its Roche lobe, heats to extreme temperatures and is rendered aglow before descending out of sight into the black hole itself.

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The stars of the Milky Way galaxy trace a big, flat circle. With a diameter-to-thickness ratio of one thousand to one, our galaxy is flatter than the flattest flapjacks ever made. In fact, its proportions are better represented by a crépe or a tortilla. No, the Milky Way's disk is not a sphere, but it probably began as one. We can understand the flatness by assuming the galaxy was once a big, spherical, slowly rotating ball of collapsing gas. During the collapse, the ball spun faster and faster, just as spinning figure skaters do when they draw their arms inward to increase their rotation rate. The galaxy naturally flattened pole-to-pole while the increasing centrifugal forces in the middle prevented collapse at midplane. Yes, if the Pillsbury Doughboy were a figure skater, then fast spins would be a high-risk activity.